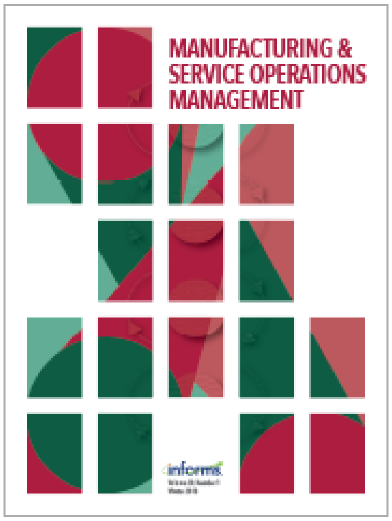


Manufacturing & Service Operations Management

Publication details, including instructions for authors and subscription information:
http://pubsonline.informs.org

Democratizing Data Analytics Products to Drive SME Growth: A Natural Experiment

Zhe Yuan, Yitong Wang, Tianshu Sun, Huilan Xu

informs
https://pubsonline.informs.org/journal/msom
MANUFACTURING & SERVICE OPERATIONS MANAGEMENT
Vol. 27, No. 4, July–August 2025, pp. 1053–1067
ISSN 1523-4614 (print), ISSN 1526-5498 (online)

Democratizing Data Analytics Products to Drive SME Growth: A Natural Experiment

Zhe Yuan,[a,b,c,d] Yitong Wang,[e] Tianshu Sun,[f,*] Huilan Xu[a]

[a] School of Economics, Zhejiang University, Hangzhou 310058, China; [b] Center for Research of Private Economy, Zhejiang University, Hangzhou 310058, China; [c] Future Regional Development Laboratory, Zhejiang University, Hangzhou 310058, China; [d] Academy of Financial Research, Zhejiang University, Hangzhou 310058, China; [e] Kogod School of Business, American University, Washington, District of Columbia 20016; [f] Cheung Kong Graduate School of Business, Beijing 100006, China

*Corresponding author

Contact: yyyuanzhe@gmail.com, https://orcid.org/0000-0002-7502-9102 (ZY); yitong_wang2019@outlook.com (YW); tianshusun@ckgsb.edu.cn, https://orcid.org/0000-0002-9786-044X (TS); xuhuilan@zju.edu.cn (HX)

Received: June 18, 2024
Revised: September 21, 2024; December 27, 2024
Accepted: April 9, 2025
Published Online in Articles in Advance: May 26, 2025

https://doi.org/10.1287/msom.2024.1138

Abstract. ***Problem definition*:** Digital platforms provide sellers with the opportunity to leverage data analytics products, granting them convenient access to statistical analyses of their business data. To promote the adoption of data analytics products and facilitate the growth of small and medium enterprises (SMEs), a world-leading E-commerce platform initiated a "Democratizing Data Analytics" (DDA) campaign in May 2021, which gives free data analytics products access to all sellers on the platform. Leveraging a unique panel data set of over 370,000 sellers and a natural experiment, our study seeks to identify the causal effect of data analytics product adoption on seller performance. ***Methodology/results*:** This natural experiment led to a sharp increase in SMEs' adoption of data analytics. Employing look-ahead propensity score matching (LA-PSM) coupled with difference-in-differences methods, we find that sellers who adopted an analytics product experienced a 14.4% increase in the number of transactions and an 18.8% increase in their revenue. We further explore the rich heterogeneity across sellers and find that younger, smaller sellers, and sellers with fewer cumulative transactions, benefit more from data analytics adoption, highlighting the value of data analytics product adoption for SME growth. Finally, we uncovered the intermediate process and mechanisms underlying such growth by examining how SMEs use data analytics and detailed operational decisions they make. After the adoption of data analytics products, sellers (1) spend more on advertising; (2) edit product titles, descriptions, and images more frequently; and (3) carry more new product stock keeping units and more product categories. ***Managerial implications*:** Our research indicates SMEs significantly benefit from adopting data analytics, underscoring the pivotal role these tools play in promoting their growth and development. Meanwhile, platforms or digital economies can increase their gross merchandise volume (GMV) by providing data analytics products to their sellers.

Funding: This work was supported by funding from National Key Research and Development Project [2024YFB3312900], the National Natural Science Foundation of China [Grants 72141305, 72192803 and 72203202], Fundamental Research Funds for the Central Universities (Z. Yuan), and the CKGSB Research Institute (T. Sun).
Supplemental Material: The online appendix is available at https://doi.org/10.1287/msom.2024.1138.

Keywords: data analytics • E-commerce • digital platform • SME growth • data-driven operations • natural experiment • PSM-DID

1. Introduction

Small and medium enterprises (SMEs) make up the majority of capital stock in the economy and are vital for economic growth and job creation in the United States, the European Union, and also China (U.S. Small Business Administration 2016, OECD 2024). However, they faced larger market fluctuations and operational challenges during the recent COVID-19 pandemic and economic downturn (Cong et al. 2024). Therefore, a critical issue of interest to both practitioners and policymakers is how to promote the growth of small businesses, as measured in both sales revenue and product variety (Carpenter and Petersen 2002). Although big data and technology are revolutionizing business (Feng and Shanthikumar 2018, Guha and Kumar 2018), one barrier for SME growth is that SMEs typically do not have the literacy, talent, resources, and infrastructure for analytics, which are required to derive value from business data beyond basic information about sales and profit. Although the adoption of data analytics products is primarily driven by large sellers, many small businesses lack the capacity to use data analytics to process business data. This discrepancy may be attributed to the relatively high costs associated with data analytics products for SMEs.

Major platforms such as Amazon, Alibaba, eBay, Facebook, Google, TikTok, and Groupon have been providing SMEs with access to data analytics products and gain value from business data (Liu et al. 2020, Zha et al. 2022). Data analytics has become increasingly important in the operations of firms (Choi et al. 2018, Cohen 2018). A range of data analytics products enables SMEs to closely monitor their daily performance, understand customer demand, track product market trends, and adjust their operations accordingly. Millions of firms now depend on these data analytics products to make store operations, product operations, and marketing operations decisions when selling their products to target audiences. However, previous literature focuses on the value of data analytics for demand forecasting (Ferreira et al. 2015, Cui et al. 2018, Cohen et al. 2022), and there is still little understanding of whether and how firms and platforms can use data analytics to improve their operational decisions and drive growth.

The scarcity of research on the value of data analytics product adoption is due to the lack of applicable empirical settings and microdata at the seller level as well as an external shock in terms of access to data analytics products. Using a unique panel data set of over 370,000 sellers and a natural experiment on a major E-commerce platform, our study seeks to identify the value of data analytics product adoption for firms' growth by exploring the causal effect of data analytics product adoption on seller performance. Specifically, by combining highly detailed data on firms' data analytics adoption, information contained in the data analytics products, detailed operational decisions, and various outcomes, we aim to answer the following research questions:

Question 1. What is the impact of data analytics product adoption on SME performance in the E-commerce platform (e.g., transactions, revenue), and why? This will help us understand the value of data analytics in driving seller growth.

Question 2. Does the effect of data analytics adoption vary across different sellers in E-commerce? This would provide important insights for policymakers and platforms on democratizing data analytics (DDA).

Question 3. What are the potential mechanisms underlying the effectiveness of data analytics? The results would show the intermediate processes of how firms can operate on their data analytics and how a digital platform should design its data analytics product to facilitate sellers' data-driven operations.

Before the launch of our collaborating platform's DDA campaign in May 2021, data analytics adoption was predominantly driven by larger sellers. However, with the introduction of the platform's free data analytics product for all sellers, a notable natural experiment occurred, resulting in a significant increase in data analytics adoption, particularly among SMEs.

Utilizing look-ahead propensity score matching (LA-PSM) combined with the difference-in-differences (diff-in-diff) method, we find that the seller adoption of data analytics products leads to a 14.4% increase in the number of transactions and an 18.8% increase in revenue (gross merchandise volume (GMV)). Additionally, we investigate the heterogeneity in the effect and find that younger and smaller sellers, and those with fewer cumulative transactions, benefit more from adoption, suggesting the value of data analytics products for democratizing SME growth.

Furthermore, we unpack the mechanisms behind these findings by analyzing SMEs' operational decisions. Specifically, we investigate the utilization of "Traffic" and "Category" analytics in detail, finding that these analytics tools benefit sellers through three channels. First, Traffic analytics support sellers' traffic operation decisions by displaying a breakdown of sellers' traffic sources and search keyword conversion rates, allowing sellers to invest more in advertisements, particularly those using keywords with higher conversion rates. Second, Traffic and Category analytics can both help sellers' product operations by revealing popular product titles (and search keywords) and popular product images, enabling sellers to modify their product titles and descriptions and change their product images more frequently. Third, Category analytics assist sellers with new product introduction and category expansion by listing the most popular products and how various categories perform. In conclusion, with the adoption of data analytics products, sellers (1) increase and better allocate their advertising spending, (2) revise their product exposition more often and better describe their products, and (3) stock more new product varieties and expand into more product categories.

Exploring how SMEs grow with the help of data analytics not only reveals the value of market data for SMEs but also provides insights into how democratizing data analytics can integrate SMEs into economic growth. Examining the value and mechanisms of data-driven growth has become increasingly important since the outbreak of COVID-19 and the accompanying economic downturn, as many SMEs are searching for new strategies to promote growth in an environment where traditional methods based on intuition or past experience no longer work. Our study aims to fill this gap.

Our research offers several contributions to the existing literature. Firstly, we examine the data gap between large firms and SMEs, and how data analytics tools can help bridge this gap. Traditional literature mainly focuses on the impact of data analytics on larger sellers, as they were the primary adopters of these tools. However, with the introduction of the platform's free data analytics product available to all sellers, there has been a significant increase in the adoption of data analytics among SMEs. This change makes the study of data analytics tool adoption by SMEs feasible. Our research shows that SMEs gain greater benefits from adopting data analytics, underscoring the vital role these tools play in promoting their growth and development. Secondly, we explore the different mechanisms by which data analytics tools impact seller performance. Whereas previous technology adoption research shows that information can decrease costs, we find that data analytics tools aid in sellers' traffic operations, advertising, and product operation decisions. This discovery is based on the functions of the product and our interactions with firm operators. Lastly, we quantify how different mechanisms drive seller growth, particularly in relation to the data analytics product. Our comprehensive data set includes the type of data that sellers can access, their operational decisions, and the outcomes of their performance, which make the exploration feasible. By probing these mechanisms, we offer valuable insights into how sellers can use data analytics to improve their operational decisions and performance.

The rest of the paper is organized as follows. Section 2 summarizes the relevant literature and contributions. Section 3 outlines the research setting and the exogenous natural experiment. Section 4 explains the empirical strategies we used to identify the impact of data analytics product adoption on seller performance. Sections 5 presents the main results and the heterogeneity. Section 6 explores the underlying mechanisms. Finally, Section 7 concludes the paper.

2. Related Literature and Contributions

Our study builds on and contributes to multiple streams of literature in operation management, information systems, marketing, and finance, including the relationship between information technology (IT) and firm performance, the value of data and data-driven decision making (DDD), artificial intelligence (AI), and the data assets in economic growth as well as the digital platform ecosystem.

The incorporation of data analytics into firms' operations stems from a longstanding tradition of IT usage. Since the mid-1990s, it's been widely recognized that information technology serves as a significant driver of firm growth (e.g., Brynjolfsson and Hitt 1996, Bresnahan et al. 2002). Sellers benefit from investing in information technology as IT adoption influences business strategies (Bartel et al. 2005), labor decisions (Bresnahan et al. 2002), and complementary organizational practices (Lichtenberg 1995, Brynjolfsson and Hitt 1996, Dewan and Min 1997). Dong et al. (2009) studied the value of information technology in supply chains. Regarding heterogeneous effects, Tambe and Hitt (2012) found that returns on information technology are substantially lower for small and midsized sellers compared with larger ones.

After the resolution of the "productivity paradox," literature began to examine the value of data and information for sellers and markets (Feng and Shanthikumar 2018), where data analytics has become increasingly crucial in firms' operations. Various data analytics tools allow SMEs to monitor their daily performance, understand customer demand, track product market trends, and adjust their decisions accordingly. Fisher and Raman (2018) demonstrate that big data analytics can aid retailers in innovating their operations and business models, as well as improving their current business model execution. Guha and Kumar (2018) examine how big data analytics has been employed in the domains of information systems, operations management (OM), and healthcare, discussing its potential and challenges. Choi et al. (2018) delve into big data analytics techniques, their strengths and limitations, strategies to address computational and data challenges, and their application in various operations management areas such as forecasting, inventory management, and supply chain management. Cohen (2018) investigates the transformative impact of big data on the service industry and several mechanisms used in the service industry to leverage the potential information hidden in big data.

In the 2010s, with the rising interest in big data, researchers continued to study the impact of technology-enabled data access empirically, focusing more intently on the effects of data analytics and DDD. A series of literature, including works by Brynjolfsson et al. (2011, 2021), Saunders and Tambe (2015), Wu and Hitt (2015), Müller et al. (2018), Brynjolfsson and McElheran (2016, 2019), and Bar-Gill et al. (2024), demonstrates that sellers making data-driven decisions have higher output and productivity. Caro and Gallien (2007) underscored the importance of demand learning in influencing product assortment decisions, demonstrating its potential to effectively shape operational strategies. Cui et al. (2015) argued that incorporating downstream sales data for order forecasting could significantly reduce the mean squared forecast error, thereby enhancing overall data accuracy. Bastani and Bayati (2019) introduced a new algorithm that tailors decisions to individuals based on high-dimensional data, outperforming current methods and physicians in a medication dosing scenario. Cui et al. (2023b) explored the trade-off in platform information provision concerning delivery speed, suggesting that such trade-offs could be used by platforms to optimize their delivery promises, thus boosting customer satisfaction and operational efficiency. The literature also emphasized information such as social media information (Cui et al. 2018), customer-related information (Zeng et al. 2020), health data (Hopp et al. 2018, Li et al. 2021b), omni-channel information (Gallino and Moreno 2014, 2018), customer engagement (Wang et al. 2022a), product assortment (Wang and Lu 2024), and sales data (Li et al. 2021a). The DDD literature also concentrates on the complementary practice of data usage, such as organization design (Key Performance Indicator in Brynjolfsson and McElheran 2016), data talent (Tambe 2014, Wu and Hitt 2015), firm-specific investments in general-purpose technologies (Tambe et al. 2020), and workplaces designed for high flow-efficiency production (Brynjolfsson et al. 2021). To further expand the DDD literature, we investigate the value of adopting data analytics products. Specifically, we analyze the effect of adopting data analytics products on SME growth, as opposed to large sellers where data are more easily accessible. Furthermore, our detailed data allow us to study how different types of market information provided by data analytics products influence firm decisions and the underlying mechanisms, which were previously under-explored because of data limitations. In summary, we complement the existing literature and show how democratizing data analytics can effectively enhance data-driven decision making.

Recently, a new strand of Operation Management and Information Systems literature has turned its focus onto the business value of AI technology (Mithas et al. 2022). Ibrahim et al. (2021) propose that human point forecasts can enhance data-based algorithm predictions when used as inputs. Bastani et al. (2021) highlight the potential of reinforcement learning and real-time data to improve public health measures. Cui et al. (2021) investigate the effect of buyers using artificial intelligence on the pricing strategies adopted by suppliers. Gijsbrechts et al. (2022) show that applying reinforcement learning significantly improves inventory management procedures. Lu and Zhang (2025) found that human involvement considerably lowered the default rate compared with decisions made solely by machines. Wang et al. (2022b) designed a deep reinforcement learning-based AI agent that learns faster and generates more revenue through personalized targeting strategies in a sequential setting. Khern-Am-Nuai et al. (2023) present results indicating that AI-based systems outperform crowd-based systems in terms of efficiency when choosing cover images. Tanlamai et al. (2024) examine the impact of AI on housing price settings, with a particular focus on the resultant consequences related to discrimination. Additionally, the literature discusses the importance of AI's complementary assets, such as human resource management (Tambe et al. 2019), employment and organizational structures (Dixon et al. 2021, Xue et al. 2022), employee investments in AI skills (Rock 2022), and customer experience (Cui et al. 2023a).

Our current research differs from prior studies on information technologies (e.g., Enterprise Resource Planning, Supply Chain Management, Customer Relationship Management, Electronic Data Interchange, AI, etc.). First, information technology is hardware and software, whereas data analytics products are one type of information technology which allows SMEs to monitor performance, understand customer demand, track market trends, and adjust operations. There are other types of information technology, such as general-purpose technology and communication technology. We focus on helping sellers access data analytics products, interpret the business data obtained, and explore customer demand and market conditions using data analytics. Second, the mechanism is different. Previous research on technology adoption demonstrates that information decreases costs, but we find that data analytics products help sellers' traffic operation, advertising, and product operation decisions. Third, and most significantly, we deconstruct the black box and illustrate the operational processes and intermediate mechanisms underlying the value of data analytics products, which are usually not viable in the previous research in OM because of the lack of detailed data on sellers' operational decisions. Our data include the type of data that sellers can access, their operational decisions, and the seller performance outcomes. Thus, we can examine the detailed change in operational decisions and unpack the impact of data analytics on firm outcomes at a granular level. Therefore, we can identify the relationship between the access to data analytics products and the firm's operational decisions.

3. Research Setting and the DDA Campaign

3.1. Importance of SMEs

This paper focuses on SMEs because SMEs are important component of the market. In fact, according to the U.S. Small Business Administration, small businesses make up about 44% of U.S. economic activity, and they create two-thirds of new jobs, driving much of the employment growth. SMEs make up the majority of capital stock in the economy and are vital for economic growth and job creation (U.S. Small Business Administration 2016). SMEs contribute approximately 50% of tax revenue, 60% of gross domestic product (GDP), 70% of technological innovation, and 80% of urban employment in China (OECD 2024). In 2022, there were about 52 million SMEs in China, and the number of newly registered enterprises reached an average of 23,800 per day (OECD 2024). SMEs promote inclusive growth by providing opportunities for entrepreneurship and self-employment.

Despite their importance, SMEs are more vulnerable compared with the large sellers. SMEs have limited access to data because of capacity constraints and often adopt fewer data analytics tools because of the substantial investment required. As a result, SMEs face larger market fluctuations (changes in consumer demand have a more significant impact on them) and operational challenges (lack of established processes and infrastructure) (Anderson et al. 2018, Cong et al. 2024).

Democratization of data analytics helps narrow the data gap. The examination of the value and mechanisms of data-driven growth has gained prominence since the advent of COVID-19 and the ensuing economic downturn. Many SMEs are actively seeking new strategies to stimulate growth in a landscape where traditional methods, grounded in intuition or past experience, are falling short. Consequently, an exploration of how SMEs can leverage data analytics for growth does not just underscore the value of market data for these businesses but also sheds light on how the democratization of data analytics could incorporate SMEs into the broader economic growth narrative.

3.2. Research Context

In order to examine the impact of data analytics adoption on SMEs, we have partnered with a globally recognized E-commerce platform. This collaboration will allow us to scrutinize how DDA influences seller performance. The collaborating platform has a huge online presence, with a large variety of products and merchants. It has over 10 million sellers and 650 million buyers. Each day there are tens of millions of transactions and billions of USD of GMV per day. As of 2018, the platform represents a two-digit percentage market share of E-commerce in China. More importantly, a large portion of the sellers on the collaborating platform are small and medium sized, which provides us an ideal environment to study the value of data analytics products to SMEs.

Most SMEs do not have the capacity to collect and process business data. According to the platform survey, most SMEs on the platform employ no more than 10 employees, and none of them are data analysts. They therefore lack the literacy, the resources, and the infrastructure to collect and access business data for their operations and extract meaningful insights from them. For instance, most sellers do not maintain their own comprehensive data assets. They can only keep track of their operational data, such as sales and product inventories, and it is challenging for them to monitor their customer traffic composition, collect information on product market demand trends, and monitor their competitors. There are third-party data analytic products available; however, they are quite restricted and not widely adopted by the SMEs.[1]

3.3. Data Analytics Products

The collaborating platform runs several data analytics products to assist SMEs in accessing and utilizing their business data. Specifically, we discuss Traffic analytics and Category analytics. These two data analytics products process seller operation status and market conditions into business intelligence. Based on these data and business intelligence, sellers can make data-driven decisions. They can monitor traffic sources with Traffic analytics, which specifically provides traffic source decomposition (advertising traffic versus organic traffic) as illustrated in Online Appendix A. This information allows sellers to make more efficient decisions on advertising expenses. Traffic analytics also lists a set of keywords with higher conversion rates and high customer demand. Sellers can better operate by employing more profitable keywords.

Meanwhile, Category analytics help sellers comprehend their products' performance and various categories' performance, and present the list of popular products and categories. Such analytics thus help sellers optimize their product operation and category entry decisions. Despite this, this data analytics tool does not offer price analysis or smart/dynamic pricing functions (Li et al. 2016). Section 6 delves into the specific functions and illustrates how these functions influence the operational decisions of sellers and aid in their growth.

3.4. DDA Campaign

The E-commerce platform designed and launched its DDA campaign in May 2021. As part of this campaign, the platform made its data analytics product available free of charge to all sellers, aiming to level the playing field and empower SMEs with the same analytical capabilities as their larger counterparts. The planning and rollout of the DDA program were not announced and are thus exogenous to the millions of SMEs on the platform, which provides a perfect natural experiment through which to determine the value of data analytics. With the DDA launch in May 2021, all sellers were able to access the platform's Traffic and Category analytics for free—these analytics products used to cost a hefty amount for each seller (the annual fee ranged from a minimum of 288 CNY to 9,888 CNY for each product, depending on the seller's number of transactions in the previous year).[2]

The DDA program was disseminated to sellers via several avenues. Firstly, it was introduced to all sellers through the "Seller Center," a tool they use daily to manage various operations such as tracking sales, managing inventory, and addressing customer service issues. Secondly, the introduction of this policy was prominently featured on several major websites; hence, it is highly likely that a majority of the sellers were aware of this information. Thirdly, given the frequent communications between sellers, information regarding the DDA program would likely have been shared and discussed among them.

As far as we know, the adoption alone does not affect the product rankings on search pages. This Democratizing Data Analytics program is a nonprofit project, and the platform provides only information to adopters to promote adoption instead of other benefits.

4. Data and Empirical Strategy

4.1. Data Analytics Adoption

To describe data analytics adoption, we have access to data analytics adoption information for a random sample of five million sellers, representing approximately 10%–30% of all sellers who have previously operated on the platform.[3] Within this data set, we are able to observe details regarding the adoption of data analytics tools by each seller, including whether they have adopted such tools, the specific type of tool adopted, and the timing of their first adoption.

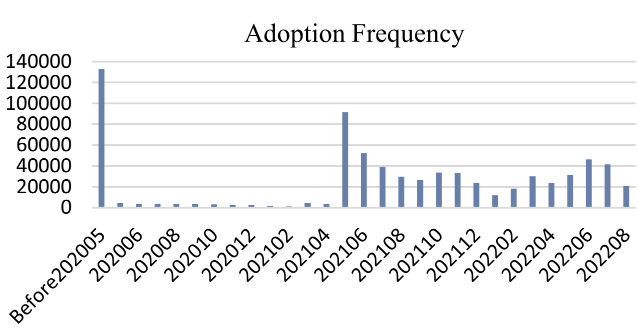
Figure 1 reports the number of Traffic and Category analytics adopters by month. The data analytics products we study were launched in 2015, but only a small percentage of sellers (around 2.6%, or 130,000 sellers) had adopted them one year prior to the DDA campaign (as indicated by the spike on the left side of the distribution). Sellers exhibited hesitancy toward adopting this particular data analytics tool, with an average monthly adoption rate of only 3,000 sellers before the campaign began. However, during the first month of the DDA campaign, there was a substantial increase in adoption rates. Approximately 100,000 sellers (91,668) adopted these data analytics products for the first time. Although the number of new adopters decreased in the following months, there was still a significant number of sellers continuing to adopt the tool.

4.2. Working Database

Because of the platform data policy, we cannot access detailed transaction and operation data for all five million sellers. As an alternative, we obtain more detailed information for a subsample of sellers who are active and have completed at least one transaction every consecutive month between September 2020 and April 2021, which leaves us 370,000 sellers. We track the performance of these sellers for eight months before and after the DDA program (in total, 16 months). These constitute our working database.

The working data set combines four pieces of information. First, we observe the sellers' data analytics adoption decisions, for instance, specific data analytics products and time of subscription. Second, our data set

Figure 1. (Color online) Number of Adopters over Time



records seller operational decisions, including advertisement decisions and the set of new products the sellers carry every month. For each product they carry, we take note of the product title, the product description, the picture identification (ID) of the main image, and the category of the product. Third, for each seller, we observe their monthly number of buyers, number of orders, and GMV. Lastly, we also compile a detailed set of seller characteristics, including seller years of operation, industry, and owner characteristics. These detailed data allow us to build a relationship between data analytics product adoption, operational decisions, and seller performance. Therefore, we can unpack the black box of how data analytics products affect the sellers' decision making and thence seller performance and illustrate the intermediate processes and mechanisms underlying the value of data analytics products. We provide a table of variable definitions and summary statistics in Online Appendix B.

Among these sellers, 71,681 adopted either Traffic or "Product" analytics before the DDA campaign. Another 105,573 sellers adopted either Traffic or Product analytics after the DDA campaign. Within these sellers, one-third (35,225) adopted either Traffic or Product analytics in the first month after it became free. Table 1 compares the characteristics of the paying users (who adopted the data analytics tool before May 2021), free adopters (who adopted the data analytics tool after May 2021), and nonadopters (who never adopted the data analytics tool). We found that, on average, paying users are five times larger than the free adopters in terms of monthly GMV and orders. Nonadopters are much smaller, about one-third the size of the free adopters. The gap in the number of years of operation is quite similar among these three groups of sellers, with the paying users having slightly longer tenure compared with free adopters.

4.3. LA-PSM

We follow prior literature in Information Systems and Marketing (e.g., Manchanda et al. 2015, Bapna et al. 2018) and employ an LA-PSM procedure in combination with the difference-in-differences method to identify the treatment effect of data analytics adoption. We match treatment group sellers with control group sellers using propensity score matching over a broad range of seller characteristics and pretreatment behaviors. By doing this, LA-PSM matches treated sellers with sellers that are currently nonadopters but will become adopters in the future. Therefore, our procedure not only accounts for observable differences but also takes into account unobserved time-invariant characteristics between control and treatment. Then, we execute a diff-in-diff analysis using the matched sample to identify the treatment effect, comparing the change in performance of the matched treatment and the control treatment before and after the adoption of data analytics products. Using PSM in combination with other parametric model estimations can

Table 1. Comparing Seller Characteristics by Data Analytics Adoption

Variable	Group name	Mean	P5	P25	Median	P75	P95	Number of sellers
Monthly GMV	1. Paying users	611,278	2,650	26,017	91,832	300,983	1,810,231	71,681
	2. Free adopters	110,866	1,122	7,564	27,420	86,136	407,887	105,573
	3. Nonadopters	32,710	355	1,576	5,516	20,355	119,748	200,373
Seller age	1. Paying users	6.8	2	4	7	9	13	71,681
	2. Free adopters	6.7	2	4	6	9	13	105,573
	3. Nonadopters	6.9	2	4	7	9	13	200,373

Notes. We restrict to sellers who are active and have completed at least one transaction every consecutive month between September 2020 and April 2021. Monthly GMV and orders are calculated between September 2020 and April 2021 in RMB.

substantially alleviate concerns about selection bias when making causal inference (Liu and Bharadwaj 2020).

The empirical approach restricts the analysis to 105,573 free adopters. One-third (35,225 out of 105,573) of sellers adopted the data analytics during the first month after Traffic or Category analytics became free (May 2021) and are defined as the treatment group. These sellers are matched with those that adopted after November 2021 (29,667 sellers), referred to as the control group.

We examine how seller monthly performance and characteristics prior to May 2021 determine seller treatment status. The set of data analytics adoption determinants consists of the set of seller-level variables used in matching, including the seller industry, seller established year, and the gender of seller owner. The set of seller month-level variables used in matching includes monthly GMV, advertisement decisions, number of products launched, number of categories operated, and percentage of good ratings prior to the DDA. We obtain the propensity score of adoption for each seller after estimation. We then use 1:1 matching and match treatment group sellers with control group sellers with a similar adoption propensity score. In other words, we match existing adopters to sellers that are currently nonadopters but will become adopters in the future. By doing this, we match sellers with equivalent observable characteristics and unobserved but time-constant characteristics.

Table 2 reports the summary statistics of the matched sample. There are 12,303 sellers in both the treatment group and the control group. We report summary statistics for five measures: monthly average GMV (transaction amount), number of active products, number of active categories, percentage of good ratings, and a dummy variable indicating whether a seller invests in advertising during a given month (advertising). For the measures we study, we find that the treatment group and control group are qualitatively similar in various measures before adoption. However, the treatment group outperforms the control group after sellers in the former category adopt data analytics products.

5. Empirical Results

We investigate the impact of data analytics adoption on seller performance, leveraging the spread of the DDA campaign. After the adoption of data analytics, sellers will be able to process their business data, gain market insights, and make data-driven decisions in areas such as traffic operation, product operation, introduction of new products, and entry into new categories. We hypothesize that sellers can draw in more buyers, fulfill more orders, and generate more GMV after gaining access to data analytics. We further examine the mechanisms through which data analytics product adoption affects seller operation decisions and eventually boosts seller transactions in Section 6.

5.1. Main Effect of Data Analytics Adoption

We employ a classic difference-in-differences empirical specification based on the matched sample. Here is the empirical specification:

$$y_{it} = \alpha + \beta \cdot Treat_i * POST_t + \omega_t + \theta_i + \varepsilon_{it},$$

where $Treat_i$ is a dummy variable which equals one if seller i adopts data analytics in May 2021. $POST_t$ is a dummy variable which equals one if month t is after May 2021. We are interested in the value of β, which represents the effect from data analytics. We further control for month fixed effect ω_t and seller fixed effect θ_i.[4]

Table 3 displays the main results. Columns (1) and (2) demonstrate significant increases of 14.4% in the number of buyers and order transactions, respectively, in the

Table 2. Matched Sample (Monthly Average)

	Pre-DDA		Post-DDA	
Outcomes	Control	Treatment	Control	Treatment
GMV	98,339	98,557	94,593	103,950
# Products	42	42	43	45
# Categories	2.3	2.3	2.3	2.3
% Good Rate	97%	98%	97%	98%
Adv Invest Dummy	41%	41%	38%	42%
# Firm-month	98,424	98,424	98,424	98,424
# Sellers	12,303	12,303		

Note. We compare five measures: monthly average GMV (transaction amount), number of active products, number of categories (seller operation scope), percentage of good ratings, and a dummy variable indicating whether a seller invests in advertising during a month (advertising).

Table 3. Main Effect of Data Analytics Adoption on Firm Performance

Outcomes	(1) ln # buyers	(2) ln # orders	(3) ln GMV
Treat × Post	0.144***	0.144***	0.188***
	(0.0124)	(0.0128)	(0.0156)
Constant	4.907***	5.126***	9.874***
	(0.00618)	(0.00640)	(0.00778)
Observations	393,696	393,696	393,696
R^2	0.009	0.007	0.009

Notes. Standard errors are in parentheses and clustered at seller level. Treat: Sellers who adopt data analytics in May 2021. Post: Equals one if month ≥May 2021. The regression is conducted on the matched sample, and no further control variables are included in the regression.

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

treatment group, as compared with the matched control sellers. In terms of order transactions, column (3) shows that the treatment group earns 18.8% more in GMV than the control group. Online Appendix C reports a series of robustness checks.

The empirical results confirm our conjecture that data analytics products can be beneficial for sellers. We observe a substantial (two-digit) increase in the number of buyers and transactions. Our improvement is higher than that identified in Brynjolfsson et al. (2011) and Brynjolfsson and McElheran (2016, 2019), who saw single-digit increases in output. There are two potential reasons for this disparity: First, we study SMEs, which are smaller sellers than those studied in the literature. As we analyze the heterogeneous effect in Section 5.3, we find that smaller sellers benefit more from data analytics product adoption. Second, we study online sellers, who face more volatile demand and are more likely to benefit from data analytics products. We conduct a mediation analysis in Online Appendix E.

5.2. Relative Time Model: The Persistent Effect of Data Analytics Adoption

As more sellers adopt data analytics products, it is possible that the benefits from data analytics adoption vanish. To verify the robustness of our results, we test a relative time model, following Greenwood and Agarwal (2015). We estimate a vector of coefficients for each month crossed with the treatment group dummy. We incorporate seller fixed effects to control for time-invariant seller heterogeneity and omit the first month in the sample as a robustness check.

$$y_{it} = \alpha + \sum_{t=DDA-7}^{DDA+7} \beta_t \cdot Treat_i * 1[Month = t] + \omega_t + \theta_i + \varepsilon_{it}$$

Figure 2 reports the estimates of parameter coefficients for GMV. The estimates of the number of buyers and number of orders are reported in Online Appendix D1. Most parameter estimates prior to DDA (May 2021) are not significant. This implies that the matching is well-behaved and the treatment and control groups were similar before adoption of the data analytics products. It also implies that our main results are not driven by the pretreatment trend. The signs after the month of DDA (May 2021) are positive and significant, illustrating the advantages of the data analytics products. The benefits appear to be persistent, as we do not observe the effect vanishing after eight months of adoption.

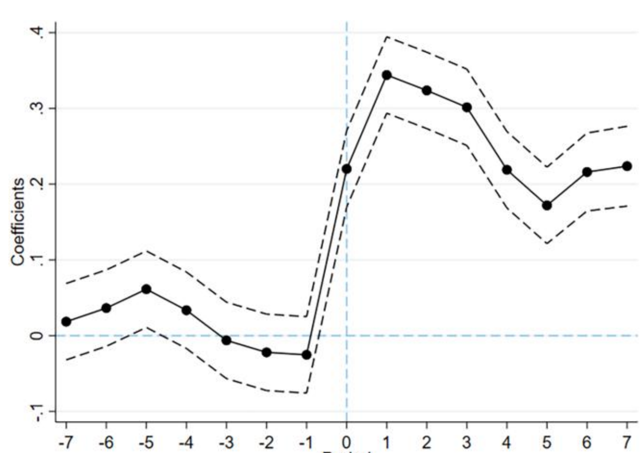
We also find that the impact is the largest right after DDA. There are several reasons that could explain the seemingly diminishing findings. Firstly, the marginal effects might decrease as sellers derive the maximum benefit when they first come across the information—going from zero to one. After several months, there are still benefits, but the impact may diminish over time. Secondly, sellers typically use the data analytics tools more frequently immediately after they adopt them. Over time, the average usage tends to decrease, which could also contribute to a reduction in the effect. We further adopt methods in Borusyak et al. (2024) and Sun and Abraham (2021) and find robust results (Online Appendix D2).

5.3. Heterogeneous Effect of Data Analytics Adoption

We further examine the heterogeneous impact of data analytics adoption. Given that our baseline is a difference-in-differences model, we employ a triple difference model, following Goldfarb and Tucker (2011), to investigate heterogeneity across sellers. We are interested in how the heterogeneity treatment effect varies across seller age, size, and reputation. Here is the empirical specification:

$$y_{it} = \alpha + \beta \cdot Treat_i * POST_t * HTE_{Index} + \gamma \cdot POST_t * HTE_{Index} + \omega_t + \theta_i + \varepsilon_{it},$$

Figure 2. (Color online) Relative Time Model



where HTE_{Index} includes the normalized values of seller age (measured by the number of years in the market), size (measured by logarithm of historical sales revenue between May 2020 and April 2021, or GMV^{hist}), and reputation (measured by seller lifetime cumulative transaction levels). We further control for month fixed effects ω_t and seller fixed effects θ_i.

Table 4 demonstrates that younger, smaller sellers and those with fewer cumulative transactions tend to benefit more from adoption. These sellers often face budget constraints and greater uncertainty, leading to hesitancy in adopting data analytics. However, they can reap significant gains from adoption, emphasizing the value of democratizing access to data analytics products. Our findings align with Berman and Israeli (2022) and Bar-Gill et al. (2024), which find that data analytics leads to a higher increase in sales for small and medium-sized sellers.

Tambe and Hitt (2012) study much larger sellers and discover considerably lower returns on information technology for smaller sellers compared with larger ones in a data set with larger sellers. One possible explanation for the disparity between their findings and ours is that the ways in which E-commerce data analytics products and IT benefit a company may differ. According to Tambe and Hitt (2012), enterprise systems like SAP, which have substantial productivity impacts, are more commonly utilized by larger sellers (Hitt et al. 2002), indicating a large cost and an economy of scale in IT deployment. However, although SMEs often struggle to afford data analytics, the DDA campaign offers all sellers affordable access to these products. As a result, smaller sellers may benefit more from data analytics, as they previously had less access to information compared with their larger counterparts.

Table 4. Heterogeneous Treatment Effect of Data Analytics Adoption Across Sellers

Outcomes	(1) ln GMV	(2) ln GMV	(3) ln GMV
Treat × Post × Years of Operation	−0.109*** (0.0134)		
Treat × Post × ln(GMV^{hist})		−1.825*** (0.0105)	
Treat × Post × Cumulative Transaction			−0.795*** (0.0125)
Constant	9.874***	9.874***	9.874***
	(0.00777)	(0.00604)	(0.00720)
Observations	393,696	393,696	393,696
R^2	0.012	0.404	0.153

Notes. Standard errors are in parentheses and clustered at seller level. Treat: Sellers that adopt data analytics in May 2021. Post: Equals one if month ≥May 2021. Years of Operation (normalized): Number of years the seller has been operating in the market. GMV^{hist} (normalized): Total transaction volume between May 2020 and April 2021. Cumulative Transaction (normalized): We use seller star level to illustrate cumulative transactions. Firm star level ranges from 0 to 20 and is based on seller lifetime cumulative transaction orders.

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

6. Potential Mechanisms Underlying the Effect of Data Analytics Adoption

We further investigate the mechanisms through which data analytics benefits sellers. To this end, we interviewed several analytics product operators and business decision makers and thoroughly explored the data analytics products. We identified and selected these mechanisms based on the insights gathered from our interviews. Our discussions with sellers operating on the platform revealed that their primary focus lies in three areas: traffic operation, existing product operation, and new product/category operation. Sellers who adopt data analytics make data-driven operational decisions to improve their performance, based on valuable insights drawn from data analytics. We synthesized their responses and present the findings in Table 5, which outlines the relationship between data analytics product adoption and corresponding seller actions driven by data analytics. We explored detailed platform data to explore the impact of data analytics adoption on seller decision making, potentially leading to improved seller performance.

We analyze different performance outcomes and discover supportive evidence for three sets of operational decisions. Section 6.1 demonstrates that sellers optimize their traffic structure by investing more in advertising, especially using query words with higher conversion rates (benefiting from Traffic analytics). Section 6.2 reports how sellers enhance their product operation activities, for example, by changing product titles, descriptions, and images (reaping the rewards from both Category analytics and Traffic analytics). Section 6.3 reveals how sellers introduce more new products and widen their product categories (benefiting from Category analytics).

6.1. Mechanism: Traffic Operation and Advertisement Decisions

Sellers monitor their customer traffic through the indices displayed in Traffic analytics and utilize it to make operational decisions about advertisement expenditure. Firstly, Traffic analytics assist sellers in comprehending their traffic sources. Specifically, sellers can track the number of visitors to their stores through different channels. When sellers notice a decline in organic traffic, they may invest in more advertisements to balance out their customer traffic. When a seller purchases an advertisement slot, their products will show up in the advertisement slots. As a consequence, this will boost the traffic to those products. On the other hand, regarding organic traffic, a seller's advertisement does not directly influence the rankings of the organic slots. However,

Table 5. Market Data and Firm Actions Based on Data Analytics

Analytics	Data	Action	Objective
Traffic	Traffic source decomposition	Invest in advertising	Traffic operation
	Keyword conversion rate	Advertise using keywords with higher conversion rates	
	List of popular keywords	Revise product titles/description	Product operation
Category	Product name recommendation	Revise product titles/description	
	Hot keyword	Revise product titles/description	
	Images of popular products	Update product images	
	List of popular products	Introduce new products	New product/category
	Performance of various product categories	Expand category	

when sellers accumulate more transactions, their ranking in the organic search might also be affected.

Secondly, these analytics also help sellers make advertising decisions more effectively by furnishing sellers with more popular query words, or those with higher conversion rates (click-through rates (CTR)). Based on this information, sellers can make more effective advertising decisions and select sets of query words for advertisements. To link these data analytics functions to our data, we anticipate that the adoption of Traffic analytics will improve seller performance by encouraging advertising that utilizes query words with a higher conversion rate.

Table 6 displays traffic operation and advertising decisions after the adoption of data analytics. Column (1) reveals that sellers are more likely to invest in advertisements after embracing data analytics. We further divide seller search queries into two groups: queries with high CTR and queries with low CTR. Columns (2) and (3) present seller advertisement page views and transactions, respectively, for search queries with high CTR. Columns (4) and (5) describe advertisement page views and transactions, respectively, for search queries with low CTR. It has been observed that sellers exhibit a substantial increase of 26.1% in advertising expenditure when targeting queries associated with higher CTR in comparison with their counterparts in the control group. However, a divergent pattern emerges when considering low-CTR queries, as treatment group sellers allocate a relatively lower increment of 22.2% in advertising investment for such queries. This is proof that sellers are willing to invest more in advertisement after data analytics adoption and data analytics aid sellers in targeting customers more efficiently (higher-CTR queries). A two-sample *t*-test analysis shows a significant difference between 26.1% and 22.2%. Columns (6) and (7) report the fraction of advertisement product views (PV) and GMV in queries with high CTR. We find that sellers receive more attention from high-CTR queries.

The absence of data analytics products leaves sellers without the knowledge and capability to effectively select keywords and allocate advertising resources. This lack of targeting ability leads to inefficient resource allocation for sellers and diminishes their growth potential. However, our research demonstrates that when provided with access to a data analytics tool, sellers gain a better understanding of the business data. As a result, they allocate more resources toward advertising, particularly on queries recommended by the data analytics tool. From a rational perspective, sellers increase their

Table 6. Effect of Data Analytics Adoption on Traffic Operation and Advertising Decision

Outcomes	(1) Ads Invest Decision	(2) Log Ads PV (high-CTR queries)	(3) Log Ads GMV (high-CTR queries)	(4) Log Ads PV (low-CTR queries)	(5) Log Ads GMV (low-CTR queries)	(6) % Ads PV(high-CTR queries)	(7) % Ads GMV (high-CTR queries)
Treat × Post	0.1264***	0.261***	0.117***	0.222***	0.0823***	0.00467***	0.00212
	(0.00537)	(0.0182)	(0.0110)	(0.0162)	(0.00932)	(0.00116)	(0.00609)
Constant		2.107***	0.554***	1.838***	0.420***	0.636***	0.608***
		(0.00912)	(0.00545)	(0.00812)	(0.00464)	(0.000393)	(0.00464)
Obs.	393,696	393,696	393,696	393,696	393,696	158,185	60,312
R^2	0.001	0.001	0.002	0.002	0.001	0.712	0.390

Notes. Standard errors are in parentheses and clustered at seller level. Column (1) reports marginal effects derived from fixed effect logit estimates. Please refer to Online Appendix F for the logit and ordinary least squares (OLS) estimates. Other columns are estimated using the linear regression model. Treat: Sellers who adopt data analytics in May 2021. Post: Equals one if month ≥May 2021. Ads Invest Decision: Equals one if a seller purchases an advertisement slot on the platform in a given month. PV: Product views. GMV: Gross merchandise volume, that is, transaction volume in RMB. High-CTR queries: Queries with click-through rate higher than the median-CTR query. Low-CTR queries: Queries with click-through rate lower than the median-CTR query.

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

investments in advertising and subsequently sell more products, as they perceive it to be a more profitable endeavor.

The increase from seller advertisement expenditure may also benefit our collaborating platform, which cares about advertisement revenue. Unfortunately, because of data limitations, we only have access to information regarding seller advertisement investment decisions and the specific queries they choose to invest in. We do not have access to detailed information on the cost of advertising or the seller's marginal cost. Further research is necessary to understand the precise impact of costs and marginal costs on seller decision-making processes.

6.2. Mechanism: Product Operation

Sellers can gain knowledge on how to operate their existing products from data analytics. Product operation is a crucial factor influencing seller product transactions in organic search. In the online marketplace, there are two stages before a transaction. First, the search engine pairs potential customers with products. Whenever a customer searches for a query word, the search engine will match the query word with products whose titles contain similar keywords. Hence, product title selection is an essential determinant of product visibility to customers. Second, sellers should write descriptions and design visually appealing images of the products, to tempt potential customers to explore these products and make transactions. Consequently, product descriptions and images are crucial determinants of customer conversion.

Sellers can optimize their products after obtaining information from Category and Traffic analytics. Traffic analytics will show a list of popular search keywords, which imply markets with higher potential demand. Based on this, sellers can modify their product titles and descriptions to better explain their products and match them with popular markets. When a product title is associated with popular keywords, sellers will receive more organic traffic from search engines.

The Category analytics offer more direct advice on product operation. For each product sold by the seller, the Category analytics provide functions like "Name Recommendation" and "Hot Keyword." Name Recommendation recommends potential revisions to product titles to assist the search engine in comprehending the title and underlying features of a product. Hot Keyword suggests trendy keywords to sellers. Sellers can revise their product titles and incorporate hot keywords into them and into their product descriptions. Category analytics also display the top-selling product in the market. Sellers can revise their product images after learning about the visual properties of the top-selling product. All these operational activities may increase customer clicks and eventually the chance of purchase.

Table 7 investigates the mechanisms through which data analytics adoption affects seller product operation decisions. The first three columns are dummy variables indicating whether sellers modify product title, description, and image, respectively. Columns (4) and (5) report the logarithm length of title and description. Columns (6) and (7) report the seller's PV and GMV from organic search.

Columns (1), (2), and (3) demonstrate that sellers are more likely to modify their titles (by 1.83%), descriptions (by 5.77%), and images (by 2.81%) after the adoption of data analytics. Columns (4) and (5) reveal that the average title length and description length are longer (by 2.2% and 1.35%, respectively) after data analytics adoption; data analytics suggests more information should be included in the product title and description, resulting in an increase in title and description length.

Table 7. Effect of Data Analytics Adoption on Product Operation Decisions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Edit dummy			Log(length)		Organic search	
Outcomes	Title	Description	Image	Title	Description	Log(PV)	Log(GMV)
Treat × Post	0.0183***	0.0577***	0.0281***	0.0227***	0.0135***	0.170***	0.188***
	(0.00426)	(0.00514)	(0.00426)	(0.00231)	(0.00317)	(0.0126)	(0.0156)
Constant				3.140***	6.018***	6.818***	9.873***
				(0.00115)	(0.00159)	(0.00629)	(0.00778)
Observations	393,696	393,696	393,696	393,696	393,696	393,696	393,696
R^2	0.000	0.001	0.000	0.001	0.000	0.011	0.009

Notes. Standard errors are in parentheses and clustered at seller level. Columns (1)–(3) report marginal effects derived from fixed effect logit estimates. Please refer to Online Appendix F for the logit and OLS estimates. Other columns are estimated using the linear regression model. Edit dummy: Equals one if the seller makes changes to the corresponding product attributes. Title: Product title, which is usually fewer than 30 Chinese characters and can be found both in the search results and at the top of the product details page (when the customer clicks on the product). Description: A detailed description of the product, which can be found on the product details page. Image: The main image of the product, which can be found both in the search results and at the top of the product details page (when the customer clicks on the product). Title length: The length of the title. With a longer title, usually the search algorithm can better understand the product and better match it with customers. Description length: The length of the product description. With a longer and more detailed description, customers can better understand the product and make purchase decisions. Organic search: Customer search in nonadvertisement slots.

***$p < 0.01$; **$p < 0.05$; *$p < 0.1$.

Table 8. Effect of Data Analytics Adoption on New Product Introduction and Category Expansion

Outcomes	(1) # New products	(2) # Categories	(3) % GMV from new products	(4) % GMV from new categories
Treat × Post	2.896***	0.0604***	0.0132***	0.0033***
	(0.999)	(0.0190)	(0.0023)	(0.0012)
Constant	41.91***	2.403***	0.0543***	0.0015***
	(0.499)	(0.00952)	(0.0025)	(0.0004)
Observations	393,696	393,696	393,696	393,696
R^2	0.000	0.000	0.000	0.000

Notes. Standard errors are in parentheses and clustered at seller level. # New products: The number of new products a seller carries in a given month. # Categories: The number of categories a seller operates in during a given month.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

These product operations have a positive effect on seller performance in terms of organic search. Columns (6) and (7) evaluate the outcomes of product operation decisions. There is a 17% rise in organic search product views and an 18.8% rise in organic search GMV for data analytics adopters, compared with the matched control sellers.

6.3. Mechanism: New Product Introduction and Category Expansion

Finally, sellers regularly launch new products and widen their categories of operation. These are the most critical decisions for seller growth. Data analytics also aids sellers in making better product introduction and category expansion decisions. Category analytics provides sellers with product-level and category-level revenue information. Sellers can learn about the top-selling products in the market and the categories that attract the most attention. As a result, they have more incentive to expand their product lines and broaden their operation scope (measured by the number of categories a seller operates in).

The results in Table 8 confirm our conjectures. Column (1) reports the number of new products carried by the seller. Column (2) reports the number of categories the seller enters. We find that sellers carry more new products and expand into more categories after data analytics adoption. Given that an average seller carries 45 products and operates in 2.3 categories, there is a 6.4% increase in the number of product offerings and a 2.6% increase in the number of categories. Sellers learn from data analytics the more profitable product and broaden its category. The increase in the number of new products and number of categories contributes to the seller GMV increment. Column (3) indicates that, following the DDA, the GMV from new products as a fraction of the total seller GMV increases by 1.3 percentage points. Similarly, column (4) reveals that the GMV from new categories as a fraction of the total seller GMV sees an increase of 0.3 percentage points post-DDA. These findings collectively suggest that an increase in both stock keeping unit (SKU) numbers and the breadth of categories contributes to the growth in revenue.

In short, Sections 6.1–6.3 summarize three mechanisms of how data analytics products benefit SMEs on the E-commerce platform. First, SMEs are not experts in advertising decisions, but data analytics products can enhance SME traffic operations and lead them to spend more on advertising (Section 6.1). Second, SMEs can learn how to better operate their products by learning from popular products and product images in the market (Section 6.2). Third, SMEs need more market trend information. They are willing to introduce more products and explore more categories after gaining access to data analytics products (Section 6.3). All these data-driven decision-making processes and improvements would help sellers gain better market outcomes.

7. Conclusion

Data analytics products provide detailed metrics on customer traffic, product demand, and market intelligence, but data analytics adoption was predominantly driven by larger sellers. We investigate a unique panel data set of over 370,000 sellers and carry out a natural experiment on a leading E-commerce platform to evaluate the value of data analytics products for SME performance. Utilizing LA-PSM along with the difference-in-differences approach, we discover a significant increase in SME data analytics adoption. A seller's adoption of a data analytics product results in a 14.4% increase in transactions and an 18.8% increase in GMV.

Our study has important managerial implications for the sellers. It sheds light on the value of data analytics products for SME growth, as well as the underlying processes. Sellers benefit from utilizing data analytics product adoption and data-driven decisions to enhance their business performance. In terms of advertisement decisions, sellers can rely on data analytics products and invest more in advertisements in potentially profitable markets (queries). Concerning product operation, sellers can learn about product information for highly sought-after products (such as product attributes, titles, and images) and the top keywords. Sellers can refine their product titles, descriptions, and images to draw in customers. Lastly, sellers can use data analytics to monitor the performance of popular products and of various product categories. They can use these data analytics products to optimize their new product introduction and category expansion decisions.

Sellers can benefit from adopting data analytics products even though these products are not free. Given that a median seller in our sample has a monthly GMV of 27,420 CNY (or $43,000 USD), a back-of-the-envelope calculation shows that an 18.8% increase in GMV is equivalent to around 62,000 CNY annually. This substantial increase in sales brings a lot of value to both the seller and the platform and is much more valuable than the cost of data analytics products.

Our findings also have significant policy implications for platforms and policymakers. A platform or digital economy can raise its GMV by offering data analytics products to the sellers on it. More importantly, we discover heterogeneous treatment effects among sellers: Sellers that are younger and smaller and have fewer cumulative transactions benefit more from adoption. These sellers previously had less business information and were more hesitant to adopt data analytics products. Platforms and policymakers can subsidize these sellers with free data analytics products and also cultivate the data analytics skills of these SMEs to sustain the development of new and small sellers and hence sustain the growth of the entire platform and digital economy.

Finally, we would like to discuss the caveats of our study. Firstly, we have primarily focused on the benefits derived from data analytics products, without delving into the details of the cost side associated with acquiring and utilizing such data. For instance, platforms bear the costs of collecting, storing, processing, and presenting business data whereas sellers incur expenses related to hiring data talent, conducting additional advertising, operating existing products, and introducing new products. Secondly, the diverse nature of sellers may obscure the fact that some sellers could experience negative outcomes when using data analytics products. Third, the provision of data analytics and the amount of data to provide are determined by the platform. The performance of sellers and market outcomes may vary depending on the data provided by sellers. Therefore, the value of data analytics is also contingent upon the platform's data provision. Lastly, our study primarily focuses on the GMV of sellers as the dependent variable because of the absence of detailed information regarding their cost structure, which includes operating costs, logistics costs, product costs, and advertising costs. This aspect can be crucial in scenarios where all sellers elevate their advertising expenditure, leading to an increase in advertising costs. In such a situation, even with an increase in revenue, there may be a decrease in profit. We leave it for future study when more detailed cost structure information becomes available.

Endnotes

[1] The data analytics tools we study in this research represent the primary and prominent offering provided by the platform. Although third-party data analytics tools are indeed accessible, they do not enjoy the same level of usage as the data analytics tool offered by the platform. This is primarily due to the platform's data analytics tool offering more precise and comprehensive information, in addition to being competitively priced. The other third-party data analytics, such as "Laihama" and "Dianzhentan," primarily focus on managing the seller's own store data and offer less comprehensive business data compared with the platform's data analytics product. These third-party tools are not as widely adopted by sellers, especially SMEs. Therefore, sellers who had not initially adopted the platform's data analytics tools, but chose to do so following the introduction of DDA, are highly unlikely to transition to third-party data analytics tools.

[2] The exchange rate is 1 CNY = 0.16 USD in 2019. In our study, sellers who adopt data analytics after DDA have a median monthly GMV of RMB 27,420, equivalent to approximately $43,000 USD annually. Assuming a profit margin of 10%, a $400 average annual fee would account for 1/10 of their total profits, which is a significant amount. It is worth noting that the financial impact can be even more substantial for sellers with lower revenues, particularly those operating in niche or long-tail markets. For such sellers, the cost of a paid data analytics product may represent a much larger proportion of their overall profits, making it potentially less viable or appealing for them to invest in.

[3] We do not report the exact number because of platform data policy.

[4] The results are accurate subject to the following two assumptions: First, we deploy an LA-PSM approach which allows us to match early adopters (treatment group sellers) with later adopters (control group sellers). Second, the data should satisfy the stable unit treatment value assumption (SUTVA). We carried out a series of robustness checks and discovered that the results remained robust, supporting the first assumption. Moreover, there are millions of sellers on the platform, yet only a small percentage of them utilize data analytics tools. This circumstance should serve to validate the second assumption.

References

Bapna R, Ramaprasad J, Umyarov A (2018) Monetizing freemium communities. *MIS Quart.* 42(3):719–736.

Bar-Gill S, Brynjolfsson E, Hak N (2024) Helping small businesses become more data-driven: A field experiment on eBay. *Management Sci.* 70(11):7345–7372.

Bartel AP, Lach S, Sicherman N (2005) Outsourcing and technological change. NBER Working Paper No. 11158, National Bureau of Economic Research, Cambridge, MA.

Bastani H, Bayati M (2019) Online decision making with high-dimensional covariates. *Oper. Res.* 68(1):276–294.

Bastani H, Drakopoulos K, Gupta V, Vlachogiannis I, Hadjichristodoulou C, Lagiou P, Magiorkinis G, Paraskevis D, Tsiodras S (2021) Efficient and targeted COVID-19 border testing via reinforcement learning. *Nature* 599(7883):108–113.

Berman R, Israeli A (2022) The value of descriptive analytics: Evidence from online retailers. *Marketing Sci.* 41(6):1074–1096.

Borusyak K, Jaravel X, Spiess J (2024) Revisiting event-study designs: Robust and efficient estimation. *Rev. Econom. Stud.* 91(6): 3253–3285.

Bresnahan TF, Brynjolfsson E, Hitt LM (2002) Information technology, workplace organization, and the demand for skilled labor: Firm-level evidence. *Quart. J. Econom.* 117(1):339–376.

Brynjolfsson E, Hitt L (1996) Paradox lost? Firm-level evidence on the returns to information systems spending. *Management Sci.* 42(4):541–558.

Brynjolfsson E, McElheran K (2016) The rapid adoption of data-driven decision-making. *Amer. Econom. Rev.* 106(5):133–139.

Brynjolfsson E, McElheran K (2019) Data in action: Data-driven decision making and predictive analytics in US manufacturing. Rotman School of Management working paper 3422397, Rotman School of Management, University of Toronto, Toronto.

Brynjolfsson E, Hitt LM, Kim HH (2011) Strength in numbers: How does data-driven decisionmaking affect firm performance? Preprint, submitted April 24, http://dx.doi.org/10.2139/ssrn.1819486.

Brynjolfsson E, Jin W, McElheran K (2021) The power of prediction: Predictive analytics, workplace complements, and business performance. *Bus. Econom.* 56:217–239.

Caro F, Gallien J (2007) Dynamic assortment with demand learning for seasonal consumer goods. *Management Sci.* 53(2):276–292.

Carpenter RE, Petersen BC (2002) Is the growth of small firms constrained by internal finance? *Rev. Econom. Statist.* 84(2): 298–309.

Choi TM, Wallace SW, Wang Y (2018) Big data analytics in operations management. *Production Oper. Management* 27(10):1868–1883.

Cohen MC (2018) Big data and service operations. *Production Oper. Management* 27(9):1709–1723.

Cohen MC, Zhang R, Jiao K (2022) Data aggregation and demand prediction. *Oper. Res.* 70(5):2597–2618.

Cong LW, Yang X, Zhang X (2024) Small and medium enterprises amidst the pandemic and reopening: Digital edge and transformation. *Management Sci.* 70(7):4564–4582.

Cui R, Li M, Zhang S (2021) AI and procurement. *Manufacturing Service Oper. Management* 24(2):691–706.

Cui Y, Jiang Z, Li Q (2023a) Unlocking the value of real-time AI assistance: Who benefits, and why? Working paper, Cornell University, Ithaca, NY.

Cui R, Allon G, Bassamboo A, Van Mieghem JA (2015) Information sharing in supply chains: An empirical and theoretical valuation. *Management Sci.* 61(11):2803–2824.

Cui R, Gallino S, Moreno A, Zhang DJ (2018) The operational value of social media information. *Production Oper. Management* 27(10):1749–1769.

Cui R, Lu Z, Sun T, Golden JM (2023b) Sooner or later? Promising delivery speed in online retail. *Manufacturing Service Oper. Management* 26(1):233–251.

Dewan S, Min CK (1997) The substitution of information technology for other factors of production: A firm level analysis. *Management Sci.* 43(12):1660–1675.

Dixon J, Hong B, Wu L (2021) The robot revolution: Managerial and employment consequences for firms. *Management Sci.* 67(9): 5586–5605.

Dong S, Xu SX, Zhu KX (2009) Research note—Information technology in supply chains: The value of IT-enabled resources under competition. *Inform. Systems Res.* 20(1):18–32.

Feng Q, Shanthikumar JG (2018) How research in production and operations management may evolve in the era of big data. *Production Oper. Management* 27(9):1670–1684.

Ferreira KJ, Lee BHA, Simchi-Levi D (2015) Analytics for an online retailer: Demand forecasting and price optimization. *Manufacturing Service Oper. Management* 18(1):69–88.

Fisher M, Raman A (2018) Using data and big data in retailing. *Production Oper. Management* 27(9):1665–1669.

Gallino S, Moreno A (2014) Integration of online and offline channels in retail: The impact of sharing reliable inventory availability information. *Management Sci.* 60(6):1434–1451.

Gallino S, Moreno A (2018) The value of fit information in online retail: Evidence from a randomized field experiment. *Manufacturing Service Oper. Management* 20(4):767–787.

Gijsbrechts J, Boute RN, Van Mieghem JA, Zhang DJ (2022) Can deep reinforcement learning improve inventory management? Performance on lost sales, dual-sourcing, and multi-echelon problems. *Manufacturing Service Oper. Management* 24(3):1349–1368.

Goldfarb A, Tucker C (2011) Online display advertising: Targeting and obtrusiveness. *Marketing Sci.* 30(3):389–404.

Greenwood BN, Agarwal R (2015) Matching platforms and HIV incidence: An empirical investigation of race, gender, and socioeconomic status. *Management Sci.* 62(8):2281–2303.

Guha S, Kumar S (2018) Emergence of big data research in operations management, information systems, and healthcare: Past contributions and future roadmap. *Production Oper. Management* 27(9):1724–1735.

Hitt LM, Wu DJ, Zhou X (2002) Investment in enterprise resource planning: Business impact and productivity measures. *J. Management Inform. Systems* 19(1):71–98.

Hopp WJ, Li J, Wang G (2018) Big data and the precision medicine revolution. *Production Oper. Management* 27(9):1647–1664.

Ibrahim R, Kim SH, Tong J (2021) Eliciting human judgment for prediction algorithms. *Management Sci.* 67(4):2314–2325.

Khern-Am-Nuai W, So H, Cohen MC, Adulyasak Y (2023) Selecting cover images for restaurant reviews: AI vs. wisdom of the crowd. *Manufacturing Service Oper. Management* 26(1):330–349.

Li X, Grahl J, Hinz O (2021a) How do recommender systems lead to consumer purchases? A causal mediation analysis of a field experiment. *Inform. Systems Res.* 33(2):620–637.

Li J, Moreno A, Zhang D (2016) Pros vs joes: Agent pricing behavior in the sharing economy. Ross School of Business Paper no. 1298, Ross School of Business, Ann Arbor, MI.

Li Y, Lu LX, Lu SF, Chen J (2021b) The value of health information technology interoperability: Evidence from interhospital transfer of heart attack patients. *Manufacturing Service Oper. Management* 24(2):827–845.

Lichtenberg FR (1995) The output contributions of computer equipment and personnel: A firm-level analysis. *Econom. Innovation New Tech.* 3(3–4):201–218.

Liu J, Bharadwaj A (2020) Drug abuse and the internet: Evidence from Craigslist. *Management Sci.* 66(5):2040–2049.

Liu Z, Zhang DJ, Zhang F (2020) Information sharing on retail platforms. *Manufacturing Service Oper. Management* 23(3):606–619.

Lu T, Zhang Y (2025) 1 + 1 > 2? Information, humans, and machines. *Inform. Systems Res.* 36(1):394–418.

Manchanda P, Packard G, Pattabhiramaiah A (2015) Social dollars: The economic impact of customer participation in a firm-sponsored online customer community. *Marketing Sci.* 34(3):367–387.

Mithas S, Chen ZL, Saldanha TJ, De Oliveira Silveira A (2022) How will artificial intelligence and Industry 4.0 emerging technologies transform operations management? *Production Oper. Management* 31(12):4475–4487.

Müller O, Fay M, Vom Brocke J (2018) The effect of big data and analytics on firm performance: An econometric analysis considering industry characteristics. *J. Management Inform. Systems* 35(2):488–509.

OECD (2024) Financing SMEs and entrepreneurs 2024: An OECD scoreboard. OECD Publishing, Paris, https://doi.org/10.1787/fa521246-en.

Rock D (2022) Engineering value: The returns to technological talent and investments in artificial intelligence. *Brookings* (June 2), https://www.brookings.edu/articles/engineering-value-the-returns-to-technological-talent-and-investments-in-artificial-intelligence/.

Saunders A, Tambe P (2015) Data assets and industry competition: Evidence from 10-K filings. Preprint, submitted September 18, http://dx.doi.org/10.2139/ssrn.2537089.

Sun L, Abraham S (2021) Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *J. Econometrics* 225(2):175–199.

Tambe P (2014) Big data investment, skills, and firm value. *Management Sci.* 60(6):1452–1469.

Tambe P, Hitt LM (2012) The productivity of information technology investments: New evidence from IT labor data. *Inform. Systems Res.* 23(3-part-1):599–617.

Tambe P, Cappelli P, Yakubovich V (2019) Artificial intelligence in human resources management: Challenges and a path forward. *Calif. Management Rev.* 61(4):15–42.

Tambe P, Ye X, Cappelli P (2020) Paying to program? Engineering brand and high-tech wages. *Management Sci.* 66(7):3010–3028.

Tanlamai J, Khern-Am-Nuai W, Cohen MC (2024) Generative AI and price discrimination in the housing market. Preprint, submitted April 12, http://dx/doi.org/10.2139/ssrn.4764418.

U.S. Small Business Administration (2016) Frequently asked questions. (January 30), https://advocacy.sba.gov/2019/01/30/small-businesses-generate-44-percent-of-u-s-economic-activity/.

Wang Y, Lu LX (2024) The value of curated boxes: Evidence from an online omnichannel fashion retailer. Preprint, submitted April 15, http://dx.doi.org/10.2139/ssrn.4771684.

Wang Y, Lu LX, Shi P (2022a) Does customer email engagement improve profitability? Evidence from a field experiment in subscription service retailing. *Manufacturing Service Oper. Management* 24(5):2703–2721.

Wang W, Li B, Luo X, Wang X (2022b) Deep reinforcement earning for sequential targeting. *Management Sci.* 69(9): 5439–5460.

Wu L, Hitt L (2015) How do data skills affect firm productivity: Evidence from process-driven vs. innovation-driven practices. Working paper, University of Pennsylvania, Philadelphia.

Xue M, Cao X, Feng X, Gu B, Zhang Y (2022) Is college education less necessary with AI? Evidence from firm-level labor structure changes. *J. Management Inform. Systems* 39(3): 865–905.

Zeng Z, Clyde N, Dai H, Zhang D, Xu Z, Shen ZJM (2020) The value of customer-related information on service platforms: Evidence from a large field experiment. Preprint, submitted March 2, http://dx.doi.org/10.2139/ssrn.3528619.

Zha Y, Li Q, Huang T, Yu Y (2022) Strategic information sharing of online platforms as resellers or marketplaces. *Marketing Sci.* 42(4):659–678.